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Review on control systems and control strategies for excavators

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Abstract. The product performance of excavators continuously improves with industrial development. However, unsighted improvements in the performance of excavators cannot meet the emission standards. Therefore, considerable research on energy conservation, high efficiency, and accurate control strategy should be conducted. At present, power matching research is performed because of the inconvenience of energy transmission. The technical advantages of hybrid power provide a better research direction for the energy-saving research of excavators. Intelligent control algorithms, such as sliding mode control and time-delay control, are added to the control system to improve the trajectory accuracy in excavator trajectory control. The intelligent operation of excavators reduces the requirements of the operating environment and improves operation economy in remote, complex, and dangerous operation spaces. The research statuses of excavators in power matching, hybrid control, trajectory control, and intelligent autonomous operation are reviewed. Finally, six conclusions about the current research status are drawn.

1. Introduction

Construction machinery has become a major pollutant emission source that cannot be ignored in recent years. As an important part of construction machinery, the hydraulic excavator is the heavy mechanical equipment mainly used in large-scale project constructions. On the premise of ensuring the project quality, it is widely used in various engineering fields, such as road construction, and bridge erection, and effectively reduces the labor intensity of workers [1]. The limitations of the excavator's working space, visual field, and the frequent occurrence of engineering accidents, such as collapses, rollover, equipment failures, and building collapses, often lead to huge casualties [2]. Therefore, studying energy-saving control strategies and intelligent autonomous operation is vital to all excavator manufacturers. Corresponding control strategies and power systems adopted in excavators can effectively improve operational efficiency, and energy utilization and prolong the service life of machines.

The research status of excavator control strategies in four aspects, namely, traditional power matching, hybrid power, trajectory control, and intelligent autonomy, is summarized. The mechanism, characteristics, and existing problems and shortcomings of various technologies are discussed and analyzed based on the research status of the existing control system of excavators. The future development trend of excavator energy-saving control is also determined.

2. Application of energy-saving control system in excavator enterprises

Excavator manufacturers have conducted long-term research and technical research to improve

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energy conservation and consumption reduction. Since the 1980s, European and American enterprises have begun to study the application of electronic control systems in excavators. The machine was equipped with an intelligent energy-saving system, combined with the corresponding power system to complete the optimal control of excavator energy-saving control, operational efficiency, and flow matching cooperatively.

Carnegie Mellon University in the United States has developed an autonomous loading system that can perform unassisted excavation tasks at the speed of a skilled worker for several hours without encountering obstacles [3]. The automatic engine control (AEC) system was applied to the hydraulic pump servo system of excavators by Caterpillar, achieving the matching between the engine and the hydraulic pump [4]. Afterward, Dynasty software was independently developed to reveal the energy consumption rule of excavators under complex working conditions based on big data analysis of measured working conditions. A fuel-saving control strategy for engines was proposed based on state identification of the whole machine to solve the problem of adaptive power matching of construction machinery, greatly reducing the energy consumption of excavators. Komatsu has developed from an open-center load sensing system to a closed-center load sensing system (CLSS). The outlet pressure difference between the pump and the valve was detected through the load sensing (LS) electromagnetic valve to control the flow, and the power was set through the Power Control electromagnetic valve to achieve constant power control. In this approach, achieving fine operation was simplified, and the power loss and consumption were greatly reduced. Fuel consumption was also reduced by 23%. Subsequently, the Pump-engine mutual control system (PEMUC) was developed by Komatsu, realizing the precise matching of the power of the main pump and the engine under all working conditions and saving energy and reduce consumption. Since 1983, the control system of Hitachi has developed from the optimized hydraulic system to the electronic load sensing system. A system with considerable energy-saving effects using engine speed control and automatic deceleration control based on information technology (IT) was designed. Later, a complete machine control system integrating engine control (EC) and pump valve control (PVC) was developed to realize the planning and optimization of the working area of the engine and the main pump, thereby greatly reducing work energy consumption. Speed-sensing control was adopted in the electronic power optimization technology system (EPOS) of South Korea's Doosan company to detect changes in speed difference in real-time, thereby adjusting the hydraulic pump displacement and improving the work efficiency of excavators [5].

In 2007, a hydraulic excavator with a positive flow control system was successfully developed by Sany in China. Compared with the hydraulic excavator with a negative flow control system, the hydraulic excavator with a positive flow control system increases its work efficiency by 8% and reduces energy consumption by 10%. The dynamic optimization power matching control system (DOMCS)and slight load sensitive system (SLSS) were successfully developed and applied to the Sany excavators, improving the energy utilization rate and operational efficiency of the excavator. In 2016, a hybrid power system based on a new boom structure and hydraulic energy storage unit was developed by Sunward Intelligent. The technology, such as collaborative optimization control of energy recovery and operation performance, was mainly broken through. Thus, the energy-saving effect and operating efficiency of the whole machine were improved while solving the reliability problem of the energy-recovery hydraulic hybrid energy-saving system. PQ adaptive real-time adjustment technology on the power matching electrical system was adopted in the XE900C developed by XCMG Group, which can minimize power loss. Then, a heavy-duty power matching control system was studied, effectively solving the inability to excavate and the power-increasing function under heavy-duty conditions, and increasing the efficiency by 8.5% [1].

At present, the electrical system of the construction machine control system is gradually being replaced by an advanced control network. The development and application of computer technology and microelectronics technology allow the excavator control system to integrate gradually new technologies, such as electronic computers, communication, and automatic control. The electronic control system of excavators with the above technologies is gradually developing from the conventional energy-saving control system to the direction of trajectory control and intelligent robots.

3. Power matching

Power matching makes the outlet pressure and hydraulic pump flow consistent with the pressure and flow of working devices as much as possible. It aims to satisfy the absorption power of the hydraulic pump matching the output power of the engine in real-time and minimize the pressure loss along with the hydraulic system and overflow, oil return, throttling, and other energy loss of the excavator.

3.1 Power matching under different working conditions

Given the needs of engineering operations, diversified objects, varied operating conditions, and environments, and drastically shifty load, the power matching under different working conditions must be adopted to meet the needs of different operating conditions and objects and to fully utilize the power and economy of the engine.

Different power mode controls were adopted by the research team to meet the needs of different working conditions of excavators. The rotation speed-sensitive control was used in changing the hydraulic pump displacement to adjust the absorption power of the hydraulic pump, and the incremental proportion integration differentiation (PID) control was applied in the power matching between the engine and variable displacement pump to reduce speed fluctuations and improve energy utilization [6]. However, overshoot oscillation of the control system and time-consuming, laborious, and difficult tuning of the PID parameter easily occurs when the deviation is large or cannot be eliminated in a short time. Yang et al. [7] proposed a control strategy combining working conditions and hydraulic pump power adaptive adjustment. A fuzzy control algorithm was adopted to realize the strategy. The control system can realize the power matching and adaptive load change of the engine hydraulic pump by using this strategy. The effect of energy-saving and consumption reduction was remarkable. However, the control system cannot achieve high precision control. Jin et al. [8] proposed the self-tuning fuzzy PID control technology to solve the problems mentioned by combining fuzzy theory and PID control with expert experience. It was applied to the excavator control system to realize the power matching between the pump and engine. Simulation and experiments showed that compared with the traditional PID regulation, fuzzy PID control improves the dynamic and static performance of the controlled process and possesses good robustness to time-varying parameters. The control effect of the controller was greatly affected because of the large pressure fluctuations. Biet al. [9] proposed to adopt back propagation neural network (BPNN) PID control in the excavator boom system. In the co-simulation analysis of Amesim and Simulink, the time of rising and stability was shortened by nearly 50%, and the signal processing lag was shortened by approximately 20%. The control precision was high, and the system had small overshoot and oscillation. However, falling into the local minimum can quickly occur. Lu [10] proposed a weight fuzzification FCM model-based fuzzy RBF neural network algorithm based on the power matching of hydraulic pump and engine to solve the problem mentioned. The simulation and real vehicle test verified the reliability and superiority of the algorithm, which exhibited strong generalization ability, high accuracy, and fast convergence speed. He et al. [11] proposed the power matching control strategy of constant power and variable power coordinated control for different working conditions. The experiment of the whole machine showed that the speed fluctuation range was small, the energy utilization rate and working efficiency were improved, and the fuel consumption was approximately 5%.

In summary, the existing control strategies have realized the power matching of excavators under different working conditions from some aspects. These strategies can also achieve certain energy savings and consumption reductions. However, each has its advantages and disadvantages.

Power matching based on different working conditions can achieve certain energy savings and consumption reductions in traditional hydraulic excavators. The working condition distribution needs the operator’s own choice. Wrong selection can easily occur because of personal factors, thereby increasing the energy consumption instead of decreasing it. Power matching based on different working conditions also ignored the oil consumption waste caused by the inconsistent power demand of excavators in different working stages, such as rotary, mining, and unloading. Finally, most control strategies in the above research were adopted to

control the hydraulic pump in the constant power region to achieve power matching. This approach was prone to interference in the engine pump and pump load matching and ignored the inner characteristics of the pump. For instance, violent torque fluctuations occur once the constant power region was abandoned. Therefore, scholars have performed research based on staged power matching to solve the above problems.

3.2 Power matching based on stages

Gao et al. [12] identified each working stage of the excavator by load and proposed a maximum acceleration operating point switching strategy to achieve switching between different working stages. They also adopted the cooperative control strategy of torque and speed feedback to achieve power matching of the excavator, thereby improving the working efficiency and reducing the fuel consumption per bucket by3.8%. Peng [13] used DAGSVM to identify the working stage of excavators and proposed a control strategy that combined speed and torque induction control to stabilize the engine operating point. An energy-saving flow control rotary system was also designed to achieve complete excavator power matching. Compared with the traditional system, this rotary system saved more than 20% of energy and realized the flow of the hydraulic pump matching with the flow of the rotary motor. However, other forms of operations, such as crushing and land leveling, were ignored, and the identification of excavation stages by an arbitrary operator cannot be achieved accurately.

4. Hybrid technology

The application of hybrid power technology in automobiles is mature. However, it is still in the research and development stage in the field of the excavator. Hybrid power systems generally used multiple power sources and integrated energy recovery and storage systems to adjust power output. Power sources provided power for the entire system independently or in combination with other sources. They can also recover and store excess energy to save energy and reduce consumption. According to the difference in energy sources, hybrid excavators can be divided into oil–electric hybrid and oil–liquid hybrid.

Hybrid systems usually use diesel engines and motors as power sources to drive the excavator. Motors are used as auxiliary power sources to convert surplus energy into electrical energy for storage. In general, batteries or supercapacitors (SCs) are used as energy storage. The hybrid excavator uses the engine as the main power source and the hydraulic accumulator (HA) as the energy storage device (ESD). The energy storage can directly release the stored energy into the hydraulic system when the energy output of the main power source is insufficient. HAs also release energy from HAs to motors.

The most critical problem of the hybrid power system included the design and adoption of a suitable control strategy to achieve the energy-saving and consumption reduction of the excavator and maintain system stability [14]. The control strategy of the hybrid power system mainly aimed to solve the system energy management problem, that is, to consider the characteristics of the main power source, auxiliary power source, and system load in the excavator comprehensively and optimize the power output of the main and auxiliary power sources of the hybrid power system. The allocation ratio can achieve the goal of energy-saving and consumption reduction of the whole machine system while keeping the state of the hybrid power system stable. The current research on hybrid power control strategies can be divided into rule-based hybrid power control, fuzzy control, and optimized hybrid power control strategies.

4.1 Rule-based hybrid power control

The rule-based hybrid power control strategy aims to determine a series of estimated working states of the hybrid power system based on actual engineering experience and theoretical analysis and divides the working area based on certain rules for the estimated working states, allowing the working area of the system to be judged based on the critical working points and to select the corresponding control mode [15].

As early as 2004, Komatsu Company took the lead in proposing a constant working point control strategy based on different working conditions and applied it to the world’s first parallel

hybrid excavator. This strategy effectively controls the engine to work in a high-efficiency area. Compared with traditional excavators, the hybrid excavator machine had certain energy saving and consumption reduction. However, the influence of the battery SOC on the whole machine system was not considered [16]. Xiao [17] proposed a control strategy of constant operating point, dual operating point, and dynamic adjustment of engine operating point based on hybrid power system structure and working condition analysis. The effectiveness of this strategy was verified by experiments. Chen et al. [18] proposed a rule-based hybrid power system control strategy, which can ensure that the engine worked in a high-efficiency area and the SOC value of the supercapacitor was maintained within the allowable range. A two-level SOC constraint was also used to ensure capacitor stability. Experiments proved the effectiveness of the above strategy. The fuel economy applied to hybrid excavators was superior to that applied to traditional excavators. This strategy was only limited to qualitative analysis and the effect of engine output power was ignored. Xu et al. [19] proposed an instantaneous minimum fuel consumption control strategy, which comprehensively considered the influence of fuel consumption and determined the ideal operating point, to solve the problem mentioned.

The control strategy based on rules was relatively simple. The engine worked in the high-efficiency area, and the efficiency was high. However, the normal working conditions of the excavator were complex, the parameters were nonlinear, and the changes were large. Such strategies had relatively fixed working points and were difficult to deal with the working conditions effectively. The most important thing was that this type of strategy did not consider the overall efficiency, the energy conversion efficiency was low, and the SOC value fluctuated frequently. Thus, the actual overall system efficiency did not reach the optimal value despite the existence of a certain amount of energy-saving and consumption reduction.

4.2 Fuzzy control

Fuzzy control is based on expert experience. After data processing, several fuzzy control rules are formed, imitating the fuzzy reasoning process of the human brain, making fuzzy decisions, and executing specific actions without requiring precise mathematical models.

Dai et al. [20] established Recurdyn and Simulink models to analyze the energy recovery of the boom by analyzing the principle of energy recovery. They also used fuzzy PID to control the engine speed. The simulation results showed that the energy recovery rate is greatly improved. Lai et al. [21] proposed a fuzzy logic control (FLC) algorithm for hybrid excavators based on the state of the engine switch. The current load state, battery SOC, and the duration of the previous engine working state were used as trigger conditions to form a parallel or pure electric drive system mode. The results verified that the FLC algorithm has strong applicability to the change of external load, and the engine switch state can improve the energy conversion efficiency. Wang [22] proposed an FLC strategy based on a genetic algorithm to optimize the membership function. Compared with the rule-based control strategy, fuzzy control fuel economy was improved by 4.1%, and the efficiency was enhanced. The engine power could not track the required power of the hybrid excavator system in real-time because of the lag of accumulator status feedback. The control strategy could not easily achieve real-time control because of the off-line optimization control characteristics of the genetic algorithm itself. Moreover, vehicle experiments were not performed to verify the effectiveness of the strategy. Zhou et al. [23] designed a fuzzy controller based on membership function optimization and torque prediction to solve the problem mentioned. They used a genetic algorithm to optimize and adjust the membership function and BP neural network to realize the demand torque prediction of the internal combustion engine. Numerical experiments verified that the torque and fuel consumption of the internal combustion engine with this controller was greatly reduced. However, the fuzzy control could only achieve suboptimal power distribution.

The fuzzy control strategy was very suitable for systems with large lags and strong nonlinearities. Its advantage was the effective use of the fuzzy connection of the control laws based on expert experience to deal with control rules that cannot be described by parameters and to find the control parameters suitable for the controlled system with different response characteristics. Thus, this strategy could optimize the engine operating point and the energy distribution and achieve good robustness. However, it was based on expert experience and

required rich engineering experience to determine the fuzzy rules, and the working conditions of the excavator were complex. Thus, the control law of fuzzy control could not adapt to the working conditions of the excavator. Moreover, ensuring the stability of the battery SOC value was difficult. Thus, most scholars combined this strategy with other optimization algorithms to adjust its control law and optimized the whole machine system.

4.3 Optimized control hybrid power control

Kim [24] proposed a hybrid monitoring control strategy based on optimal control. This strategy consisted of an optimal strategy based on equivalent fuel consumption and a dynamic programming (DP) algorithm for power distribution. The simulation showed that the fuel optimization was nearly optimal. The optimization result of the excavator under this control strategy was 30% higher than that of the traditional excavator, and the charge retention performance of the former was excellent. It's indicated that the series-parallel hybrid structure was better than the traditional structure from the aspects of cost, fuel efficiency, and payback period [25]. A power control algorithm to achieve energy and instantaneous power balance was proposed. The simulation and experimental results verified that the fuel consumption was effectively reduced by 8%. However, the simulation results were very different from the experiment results. The excavator's frequent idling state was not considered, and the experiment under complex working conditions was not performed, resulting in the load power that is lower than the system power. P. Casoli et al. [26] provided a method to compare the layout of different hydraulic hybrid systems using a DP algorithm to define the optimal hybrid layout configuration and its optimal control strategy. Cui et al. [27] proposed a multi-objective control strategy based on torque distribution optimization to solve large torque changes and low energy efficiency. This strategy divided the work area based on load characteristics and established an energy efficiency characteristic model. Energy efficiency, fuel consumption, and torque characteristics were the optimization goals. The energy management strategy of the whole system of parallel hybrid excavators was studied. The optimization goals were improved. However, a certain system dynamics performance was sacrificed. Aiming at the problem of considering engine efficiency but ignoring the overall situation, Feng [16] established a comprehensive efficiency model of battery charging and discharging by analyzing excavator operating conditions and proposed an energy management strategy based on maximizing the comprehensive efficiency of the system. The gray predicted battery residual capacity estimation strategy and the segmented variable current charging and discharging strategy were also used. The SOC value of the battery fluctuates slightly when the overall system efficiency was optimal, thereby improving the working efficiency and durability of the battery.

The optimization control of the hybrid power system was based on optimal control theory to optimize the fuel consumption and energy efficiency of the hybrid power excavator system globally and achieve reasonable power distribution, energy-saving, and consumption reduction of the hybrid power system [16]. Optimal control was divided into global optimal and instantaneous optimal. The global optimal calculation amount was large, and the real-time strategy could only be evaluated under standard operating conditions. Moreover, the global optimal control strategy could not be directly applied in real-time under actual operating conditions. Instantaneous optimization allocated energy in real-time according to current load characteristics and control parameters to optimize energy efficiency. However, it was not optimal in the overall excavation process.

5. Trajectory control

Different operations of excavators often need a combination of actuator actions. The trajectory of the excavator must be controlled, and the digging trajectory must be tracked according to the set trajectory to achieve high efficiency and quality operation. The excavator power system is a nonlinear, large inertia strong coupling system, which requires the adaptability and robustness of trajectory control. The trajectory control mainly includes PID-combined optimization control, sliding mode control, robust adaptive control, time-delay control, and intelligent control.

5.1 PID-combined optimization control

PID control is widely used in industrial control because of its simple structure and reliable work. Zhang [28] used traditional PID control to regulate the horizontal straight excavation trajectory of excavators. The straightness error was 4.8%, but the maximum straight-line error was 112mm. Aiming at the shortcomings of the traditional PID control in trajectory control, Le et al. [29] designed fuzzy PID and fuzzy self-tuning neural network (FN) controllers to realize the trajectory track control of the circular trajectory of the small electro-hydraulic excavator. Experimental analysis showed that FN control and fuzzy PID control are superior to the traditional PID control. The cumulative and maximum errors of FN control were smaller than those of the traditional PID control, indicating the enhanced robustness of the former. Du et al. [30] combined RBF neural network and PID control and applied them to the automatic control of single bucket excavator trajectory. The simulation indicated that the combination of the RBF neural network and PID control exhibits strong adaptability and good reliability. However, the adjustment time was too long. In literary works [31-33], the improved particle swarm optimization (IPSO) algorithm, improved genetic algorithm (IGA), an improved ant colony optimization (IACO) algorithm application in the optimization and tuning of PID parameters of excavator trajectory control was developed to solve the problems of the optimization algorithm in the trajectory control, including huge error and excessively long adjustment time and stabilization time. The simulation results showed that compared with traditional PID control, these algorithms have remarkable control effects. Ye et al. [31] used IPSO to achieve the self-tuning of the bucket system position PID control parameter. The simulation verified that the trajectory error of IPSO was better than that of standard particles under the three signals of step, slope, and sine. The errors of the group algorithm and the phase margin tuning method were considerably reduced, and the stability of the algorithm and the method were high. However, no experiments were performed to verify the accuracy of the simulation results.

5.2 Sliding mode control

Sliding mode control can effectively overcome the uncertainty of the control system. It is also very robust to disturbances and parameter perturbations, particularly in the control of nonlinear systems. Research institutions considered applying sliding mode control to the trajectory control of excavators because of the nonlinearity of excavator control and large load disturbance. Hyeong et al. [34] designed and applied a trajectory tracking control system on a pneumatic excavator by using an adaptive sliding mode control algorithm and testing it on the excavator. Compared with that of the traditional PID controller, the trajectory error of the aforementioned control system was reduced by 8% under the step signal, and the error range was reduced by 10% under the sinusoidal signal. Jin [35] obtained a rotation dynamic model through system identification and used a sliding mode control observer in observing the system state to suppress the vibration of the hybrid excavator during rotation. A sliding mode controller was also designed to achieve the system's stable control and tracking control. Compared with that of the traditional PI controller, the performance of the controller was proved well in simulation and real-vehicle verification. However, the use of the saturation function instead of switching action to solve the chattering problem can guarantee only the convergence to the boundary layer. It caused steady-state errors on the sliding surface. Zhang et al. [36] established the kinematics and dynamic equations of the excavator with four degrees of freedom to reduce the influence of the chattering phenomenon on trajectory tracking control. An adaptive fuzzy sliding mode controller on this basis was designed. The controller used fuzzy adaptive sliding mode control output in replacing the switching items of the sliding mode controller to solve the problem of steady-state errors in the sliding mode surface caused by the saturation function as the switching item. A fuzzy adaptive system was established to adjust the switching gain dynamically. Its effectiveness was verified on the simulation platform. Huang et al. [37] placed a low-pass filter behind the output of the sliding mode controller to achieve high-precision trajectory tracking and eliminate chattering. After the addition of the filter, the trajectory tracking control exhibited strong robustness to disturbance, and the effect of eliminating the chattering phenomenon was improved. Feng Hao et al. [38] proposed to combine RBF neural network and sliding mode control. They applied the proposed combination to excavator trajectory control. They also

designed the RBF neural network sliding mode controller (SMC-RBF) using RBF neural network to approach the uncertain nonlinear factors of the excavator quickly. The simulation showed that the robustness of the SMC-RBF to interference and errors is higher than that of the PI controller. Moreover, the tracking control accuracy under the sinusoidal reference signal and the improved trajectory accuracy was increased by 46% and 52%, respectively, when leveling.

5.3 Robust adaptive control

Robust control and adaptive control are both proposed to solve the uncertainty problem of the controlled object. Robust control refers to the control algorithm that uses a certain controller to deal with the uncertain system. It performs well in suppressing interference and compensating for unmodeled dynamics. Adaptive control refers to the control algorithm that adjusts the controller parameters online to deal with the uncertainty of the system. It can be used to overcome the uncertain parameters in the system, but it is more sensitive to external interference and unmodeled dynamics. Professor Yao Bin from Purdue University in the United States proposed a robust adaptive control strategy by combining the two. A three-degree-of-freedom dynamic model of an excavator manipulator based on the Lagrange equation was established and an adaptive robust controller was designed [39]. A discontinuous projection method was proposed based on this controller to approximate the nonlinear gain coefficient of the valve and the nonlinearity and uncertainty of system parameters. The maximum tracking control angle error of this method in the motion control experiment of the manipulator was 3.75°. Kim [40] designed a robust controller based on μ-synthesis using a cascade control structure and used the PD controller in the inner layer to achieve approximate linearization of the excavator's nonlinear system. The layer adopted the above-designed robust controller to ensure the robust performance of the uncertain nonlinear system. Each joint node used the same second-order model. Finally, the step response and experiment proved that the controller was resistant to nonlinearity. The robust control effect with uncertainty and uncertainty reached the expectations, and the overall performance of trajectory tracking control was improved. However, the controller performed poorly under load. Vu et al. [41] established the dynamic equation of the excavator manipulator based on the Euler-Lagrange method, designed an adaptive robust controller based on the state information of the excavator system, and verified the stability of the system by using Lyapunov theory. The simulation verified that the controller with the position tracking control effect was better than the PD controller under certain conditions. However, no experiment was performed to verify validity of the simulation results.

5.4 Time-delay control

The excavator system has the characteristics of nonlinearity, uncertainty, and large disturbance. For uncertain nonlinear systems, the conventional control strategy considers feedback linearization control of the controlled object or adopts a certain method for fitting processing. Delay control involves the use of a time-delay observer to analyze the dynamic characteristics of the system without modeling. The time-delay control can compensate for the uncertainty and nonlinearity of the system through the time-delay estimation without system identification. Lee et al. [42] proposed A time-delay control with switching action (TDCSA) using an integral sliding surface (ISS). They also performed it on a 21-ton hydraulic excavator on a sloped ground of 30°, 0°, and −30°. The linear motion trajectory control experiment was verified. The comparison of the time-delay control of PD-type sliding surface with the time-delay control with switching action indicated that TDSCA using ISS had stronger robustness to low-frequency disturbance and parameter fluctuation, and higher trajectory tracking control accuracy. Moreover, the vertical error of the bucket teeth was slightly inferior to the PD-type sliding surface time lag control only on the -30° working surface. Kim [43] combined discrete time-delay control with terminal sliding mode control and designed a new discrete model-free robust controller for excavator trajectory control. Thus, the acceleration information was eliminated to reduce the influence of small noise interference. The tracking accuracy of the bucket teeth was also successfully controlled within 2cm. Kim [44] et al. designed a time-delay controller with a hold function by using the corrected error and adaptive gain to improve convergence and achieve robust control of excavator trajectory tracking. The simulations

verified that the designed controller is superior to the PID controller and adaptive inertial-related controller. However, no experiments were not performed to verify the validity and the accuracy of the simulation results.

5.5 Cooperative control

Figure 1 shows most of the abovementioned trajectory control strategies for the trajectory control of the excavator. First, the excavator rotation and the position and posture space data of the actuators of the boom, bucket, and stick are converted to the joint space through kinematic equations. This process is called inverse kinematics conversion, also called inverse kinematics solution. The opposite process is called forward kinematics conversion, also called forward kinematics solution. The inverse kinematics conversion is used to obtain the expected joint angles of the excavator's rotation and the actuators of the boom, bucket, and stick $\theta_1, \theta_2, \theta_3$ and θ_4. They are compared with the actuator rotation angles, $\theta_1^{'}, \theta_2^{'}, \theta_3^{'}$ and $\theta_4^{'}$,which are obtained in real-time from the excavator, to obtain the real-time tracking errors, e_1, e_2, e_3 and e_4. According to the real-time tracking errors and state parameters, the control law is designed to control the tracking error as small as possible. According to forwarding kinematics conversion, the actual trajectory of the bucket teeth of the excavator also achieves the real-time tracking of the desired trajectory as much as possible. However, the excavator is different because of the interference of multiple actuators and other factors. If all the actuators are controlled only by the same method, the control effect of other actuators tends to be poor when the bucket control is optimized to the best. Therefore, various research scholars coordinate and combine multiple optimization methods and apply them to excavator trajectory control to solve the accuracy problem of coordinated operation of multiple actuators.

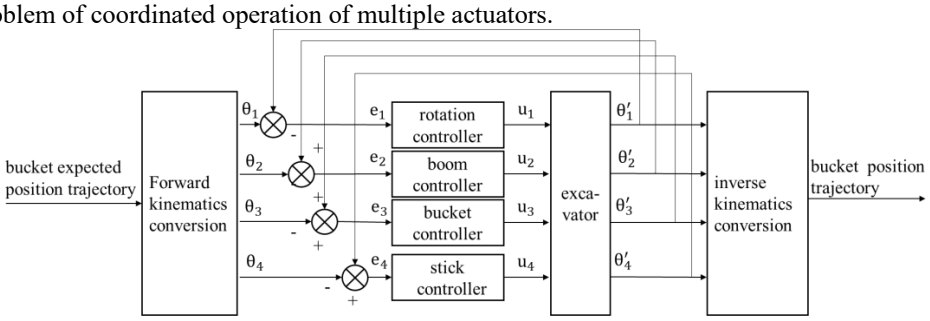


Figure 1. Bucket trajectory control flow diagram

Based on neural network approximation and terminal sliding mode control theory, Li et al. [45] combined sliding mode control, neural network, robust control, and parameter adaptive control to achieve high trajectory control accuracy of excavators. They also designed an adaptive neural network finite-time controller that used a self-learning echo state network to approximate the uncertain system function. They used a high-order sliding mode observer to estimate the uncertain state of the system and robust control to solve the uncertainty caused by the approximation and observation errors. Parameter adaptive control online learning was also used to update the upper limit of uncertainty, suppress the influence of uncertain nonlinearity without parameter identification and backstepping control and achieve an excellent control effect. Kassen et al. [46] proposed a robust feedback linearization control strategy based on sensitivity theory. They developed a comprehensive, robust optimization design controller using this strategy and the bilinear matrix inequality. It was applied to the land leveling control of the excavator bucket, and the superiority of the method in terms of robustness and energy consumption was verified. Lee et al. [47] introduced cross-coupling control into excavator trajectory control through kinematic analysis. They designed an electrohydraulic servo control system based on excavator trajectory contour control and used it on a 1.5-ton hydraulic excavator. The experiments verified that the cross-coupling control can effectively improve the accuracy of contour control by conducting leveling operations on inclined surfaces of 0°, 20°, and 45°. However, the accuracy of trajectory control was slightly reduced.

6. Intelligent autonomous control

Traditional excavators are heavy-duty, manually operated equipment. However, the difficulty and intensity of excavators gradually increase because of the rapid development in the industry. Moreover, excavators often need to work in harsh working environments, with loud noise and possible collapse accidents, which are unsuitable for manual operation. As the expectation for comfort and efficiency of excavators increases, intelligent autonomous excavators begin to develop gradually.

6.1 Automatic control

Automatically controlled excavators can use sensors to accept changes in external factors. On this basis, integrated analysis can determine the state of the excavators, allowing the excavators to respond, self-adjust, and adapt to external changes, and ensure that the entire system is in a stable operation state. As early as the end of the last century, scholars have studied the possibility of unmanned control of excavators [48, 49].

(1) Identification of soil parameters

In the early stage of excavator research, scholars believed that the online identification of excavation target soil and the online estimation of its parameters were two necessary conditions to achieve automatic control of excavators. Therefore, Yahya [50] applied the Newton-Raphson (NR) method, the generalized Newton (GN) method, and the least square method to the parameter identification of the connecting rod of the full-scale unmanned excavator. The measured data indicated that the prediction accuracy of the NR method and the GN method is high, and the calculation speed of the NR method is much faster than the calculation speeds of other methods. Thus, the NR method highly tends to be applied in practice. The GN method is more robust to noise interference than the other methods. Choo et al. [51] applied the NR method to the online real-time estimation of soil parameters. They combined the Mohr-Coulomb soil model and the Chen and Liu upper bound soil model to approximate the destructive force on the soil. This method was more robust than other methods, and its performance was better than that of other methods in sensitivity tests. He et al. [52] established an analytical model of excavation resistance by improving the analysis method of soil failure force. On this basis, they used the improved Newton iteration method and the parameter space cross method to identify the soil parameters. The simulations and experiments verified that the methods effectively identify the soil parameters. However, the maximum error value of the final identification was nearly 30% because of the strong nonlinearity of the soil itself.

(2) Automatic excavation based on trajectory planning

If automatic excavation is achieved in an excavator, real-time trajectory tracking and control with high accuracy based on the identification of soil parameters need to be achieved. The excavator controller must also convert the real-time control value of the corresponding trajectory of each working device of the excavator according to the external information received by the sensor to achieve automatic excavation under the expected trajectory of bucket excavation. Jud et al. [53] established a 2.5-dimensional excavator elevation map based on the fusion of external perception and ontology perception mapping. The sequential quadratic programming method was also adopted to select the control points of the excavator trajectory planning and a hierarchically optimized inverse kinematics controller was designed to achieve the automatic control of the excavator and the automatic excavation of the four-section plane and free-form surface. The average error was 0.027m at the maximum, which was reduced by 87%. Gu [54] successfully applied the inverse kinematics equations and proportional-integral plus to the automatic positioning control system of the excavator bucket, which can quickly and smoothly control the bucket to reach the target position and efficiently and autonomously operate. However, it was only suitable for specific excavators and was not universal. Zhong [55] used the cubic polynomial and integral method to plan the excavator's trajectory, combined the blind man's pathfinding algorithm and the preimage planning algorithm to achieve the automatic obstacle avoidance of the excavator, and adopted the cubic nonuniform rational B-spline curve trajectory planning method to achieve the timely and effective adjustment of the excavator's action in the event of a collision and ensure the continuous and autonomous smoothness of the excavator. However, the simulation and actual vehicle verification were not

conducted, and a complex nonlinear soil–tool interaction model was not built to simulate the actual mining process closely.

(3) Automatic excavation for the field environment

The automatic excavation robot oriented to the field environment must be achieved based on intelligent obstacle avoidance, automatic identification of working conditions and environment, trajectory planning, task decomposition, and soil parameter identification.

The computerized intelligent excavator was successfully developed by Lancaster University, which can automatically excavate trenches under different working conditions [54]. The intelligent mining system was developed by the Joint Research Institute by integrating the analysis of operators and other intelligence into the task planner, which can automatically make the optimal mining plan according to the three-dimensional model of the working environment. Functions such as automatic obstacle avoidance, intelligent automatic mixing, and computer-assisted remote control also existed [56]. The automatic excavation system developed by Dunbabin [57] can achieve online monitoring of the bucket full rate of the excavator bucket and the optimal excavation path planning. Yamamoto et al. [58] developed an IT execution system by combining the developed three-dimensional information mapping system, human-computer interaction interface, and excavator automatic control system to achieve the automatic excavation. The actual trajectory and the expected trajectory error were controlled within 0.1m

Zhao [59] built an excavator virtual prototype system based on OpenGL and achieved virtual automatic mining. Bi [60] proposed a two-stage method of optimizing excavation trajectory planning, established a virtual prototype model of the excavator and achieved the optimal trajectory planning simulation of the excavator. Liu et al. [61] developed an automatic control system through the hardware transformation of the excavator and the modeling and simulation of the controller. The expected angle and expected position error were controlled within 1° and 7cm, respectively. However, the control process had a certain delay.

6.2 Remote control

Given the factors such as dangerous working conditions and limited visibility of operators, remote control technology has gradually developed to ensure the safety of operators and complete excavation tasks with high quality. Many countries have also conducted in-depth research.

(1) Remote control of excavator based on wireless communication

In the 1980s, John Deere Company of the United States developed a wireless remote control hydraulic excavator with a teleoperation distance of more than 1500 m. In 1983, after the Japanese Komatsu PC200-2 excavator passed the hardware modification and test verification, it could remote control the movement of a single actuator or the compound action of multiple actuators. Hanyang University developed an unmanned remote excavator control system based on a movable manipulator in combination with environmental perception and a remote control station. However, it was limited to communication within a range of 50m [62]. Later, the remote control developed by Sunward Intelligence and Central South University achieved the remote control of excavators from 100m. The research team of Zhejiang University developed an excavator remote control intelligent system, which can achieve the functions of excavator remote driving, intelligent planning, and intelligent fault detection and diagnosis. The rapid development of wireless communication networks had made clear video images a requirement for remote control of excavators. The low latency, a wide range of support, and large bandwidth of 5G communication met the requirements for remote control of high-definition video images.

In 2018, Doosan demonstrated an excavator remote control technology based on 5G communication. It was developed for the first time, and it could remote control an excavator beyond 880km for unmanned excavation, with a response delay of only 20ms. In 2019, Shandong Lingong Construction Machinery Co., Ltd. cooperated with China Unicom to achieve the remote control of excavators based on the 5G network, real-time transmission of unmanned operation scenes, and high-definition video images [63]. Liugong Machinery Co., Ltd. cooperated with Huawei and China Telecom to develop the 9075E excavator, which could realize remote control, intelligent material identification, and task planning. The remote control excavator researched by Sany, Huawei, and China Mobile can control unmanned operations

within 10cm. Kobelco developed a 5G-based remote control system (K-DIVE) that could operate the excavator in real-time in the office. XCMG conducted in-depth research on the remote control by combining image recognition with environmental perception and achieving 5G remote control of wheeled excavators through the path, task planning, and hardware transformation.

(2) Remote control based on force feedback

Force feedback means that the interaction force between the slave excavator and the soil is transmitted to the master operating handle through the sensor. Thus, the operator can feel the excavator's on-site excavation situation to achieve operator control over the excavator with a sense of presence. Zhou et al. [64, 65] achieved real force feedback by applying DirectPlay and DirectInput technology to wireless remote control mining robots to solve the obstacles encountered by the remote control slave excavator bucket, including the inability to continue mining when the operator cannot perceive. Phantom tactile devices were applied to the remote control of excavators and a remote control system for backhoes based on simulation and tactile control was successfully developed by The Fluid Power and Motion Control Center of Georgia Institute of Technology in the United States to provide force feedback to the operator [66]. Kim et al. [67] applied the tactile device to the remote control of excavators. Thus, the operator had an immersive excavator action experience, making the operation easy. Based on kinematics and dynamics analysis, Kim et al. [68] developed a dual-side remote teleoperation control system with force feedback by using the wave variable method. The successful movement mapped the excavator bucket, stick, and boom to the operator's index finger, wrist, and elbow. Only one hand was needed to operate the equipment, overcoming the shortcomings of traditional joysticks and Phantom tactile devices that were complex and can easily cause fatigue.

(3) Remote control based on wearable devices

Japan Port and Aviation Research Institute developed a remote control system for excavator robots, which can remote operate excavating robots for underwater operations. This system greatly guaranteed the life safety of operators. In 2008, Doosan Infracore cooperated with Seoul National University to develop a remote control system for a master-slave excavating robot based on arm movements, the use of multi-sensor real-time detection of human arm movements was achieved to control the movements of excavator buckets, sticks, and booms, greatly reducing the difficulty of excavator operation, and the actual excavation error was within ±80mm [69]. However, the system needed to maintain a certain posture for a long time, easily causing fatigue, which reduced the work efficiency of the excavator to a certain extent. Thus, it could not realize the combined action of the excavator's rotation and boom lifting during the remote control of the excavator. Xiang [70] redesigned the sensor layout plan, proposed an improvement plan based on the action of the remote control of the excavator, and applied fuzzy PID to the remote control of the excavator to solve the above problems. The remote control of the excavator was verified on the simulation experiment platform and finally achieved the combined action of the excavating arm and the rotation. The control accuracy and operational efficiency were improved. Le et al. [71] designed an intelligent remote control system for excavators that combined visual feedback and a portable control station. A wearable display was used, and the real-time feedback of environmental information was achieved. This system can intelligently track the movement of the human head and switch the view direction in real-time, greatly reducing the working time of the excavator. However, the operator's field of view was blocked because of the use of a single charge-coupled device camera. Therefore, some limitations may cause mistakes in tasks to exist.

The remote control excavator has great safety advantages. However, the necessary modifications for the hardware and software of the traditional excavator are costly. Most of the remote operated excavators at this stage are small- and medium-sized excavators. In actual engineering applications, the actual operational efficiency of small- and medium-sized excavators is low. Moreover, most excavator manufacturers are still in the development stage in the field of remote control, and many technologies that need to be implemented cannot be applied to mass production.

7. Conclusion

In this review, the current research status of the excavator control system is investigated by analyzing the current research status of the control strategies of traditional power matching, hybrid power, trajectory control, and intelligent autonomy. The mechanism, characteristics, and shortcomings of each technology are discussed. The existing problems are analyzed, and the following conclusions are obtained:

(1) The traditional method of power matching according to working conditions ignores the waste caused by the different load demand power during the excavation and unloading stages of the excavator operation. A load stage identification must be established to consider the load changes in different stages of the excavator operation and achieve staged matching.

(2) Rule-based hybrid power control, optimized hybrid power control, and fuzzy control strategies in hybrid power systems mostly use heuristic algorithms to achieve power distribution, allowing the engine to work in the high-efficiency area. Among these strategies, fuzzy control is the most robust. The optimization control gradually shifts from single-objective optimization to multi-objective optimization, taking energy efficiency and optimal fuel consumption as the goals and the battery SOC value stability and torque as the constraint conditions.

(3) Trajectory tracking control is mainly based on the expected trajectory to reverse the expected angle and design the control law based on the real-time control error. It is now transformed into multiactuator coordinated control to improve trajectory accuracy, which lays the foundation for remote unmanned control.

(4) At this stage, intelligent autonomous excavators are mainly operated under standard working conditions. Industrializing the intelligent autonomous operation of excavators by combining technologies, such as environmental perception, soil parameter identification, task planning, and trajectory tracking with high-precision control algorithms, is one of the future development trends of excavators.

(5) The cost of remote controlling hardware and software reconstruction is huge. This reconstruction is mainly used in small- and medium-sized excavators and is difficult to mass-produce. The real-time performance, distance, and video quality of the remote control are greatly improved by 5G communication because of its low latency, a wide range of support, and large bandwidth. Therefore, remote control excavators based on 5G communication are also considered future development directions.

(6) The cloud service of an intelligent control system for excavators, that is, real-time data collection, big data simulation analysis, and real-time optimization of big data solutions for excavators based on simulation analysis results to adapt to various complex working conditions have become popular research directions at present. Therefore, the corresponding software will be successfully developed and applied to excavator control in the future.

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